

Lab 3 - Routing Protocol

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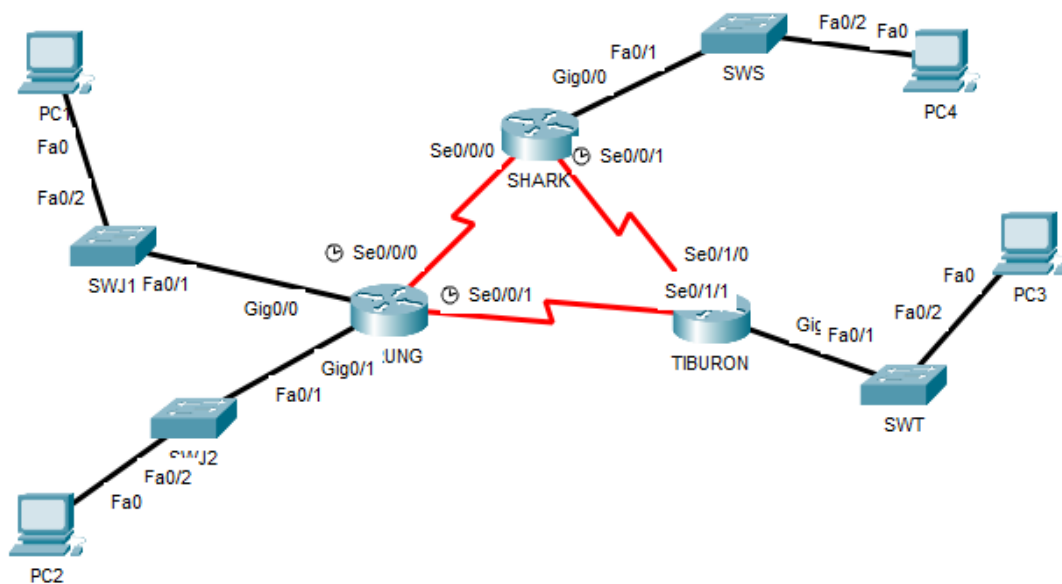
MATRIC NUM: A23CS0060

SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0(pc1)	172.18.110.1	255.255.255.224	-
4		G0/1(pc2)	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0(pc3)	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0(pc4)	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

- Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.

Device	Subnetwork	Usable Host	Usable Address	Network Address	Broadcast Address	Subnet Mask
JERUNG	LAN 1	$20=2^5=32=27$	172.18.110.1-172.18.110.30	172.18.110.0	172.18.110.31	255.255.255.224
	LAN 2	$30=2^5=32=27$	172.18.110.33-172.18.110.62	172.18.110.32	172.18.110.63	255.255.255.224
TIBURON	LAN	$50=2^6=64=26$	172.18.110.129-172.18.110.190	172.18.110.128	172.18.110.191	255.255.255.192
SHARK	LAN	$60=2^6=64=26$	172.18.110.65-172.18.110.126	172.18.110.64	172.18.110.127	255.255.255.192
Connections	JERUNG-SHARK	$2=2^2=4=30$	172.18.110.193-172.18.110.194	172.18.110.192	172.18.110.195	255.255.255.252
	JERUNG-TIBURON	$2=2^2=4=30$	172.18.110.197-172.18.110.198	172.18.110.196	172.18.110.199	255.255.255.252
	SHARK-TIBURON	$2=2^2=4=30$	172.18.110.201-172.18.110.202	172.18.110.200	172.18.110.203	255.255.255.252

- Using the IP addresses calculated, configure the devices with the appropriate information.

JERUNG

JERUNG

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.1636.2A01

IP Configuration

IPv4 Address 172.18.110.1

Subnet Mask 255.255.255.224

Tx Ring Limit 10

Equivalent IOS Commands

```
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#no shutdown
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/1
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#no shutdown
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/0
JERUNG(config-if)#
```

☐ Top

JERUNG

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.1636.2A02

IP Configuration

IPv4 Address 172.18.110.33

Subnet Mask 255.255.255.224

Tx Ring Limit 10

Equivalent IOS Commands

```
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/1
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#ip address 172.18.110.197 255.255.255.252
JERUNG(config-if)#no shutdown
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/0
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/1
JERUNG(config-if)#
```

☐ Top

JERUNG

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/0/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.193

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
JERUNG(config-if)#ip address 172.18.110.193 255.255.255.252
JERUNG(config-if)#no shutdown
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/0
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/1
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/0
JERUNG(config-if)#
```

☐ Top

JERUNG

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/0/1

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.197

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/0
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface GigabitEthernet0/1
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/0
JERUNG(config-if)#
JERUNG(config-if)#exit
JERUNG(config)#interface Serial0/0/1
JERUNG(config-if)#
```

☐ Top

SHARK

SHARK

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0040.0B2C.0A01

IP Configuration

IPv4 Address 172.18.110.65

Subnet Mask 255.255.255.192

Tx Ring Limit 10

Equivalent IOS Commands

```
SHARK(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/1
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#no shutdown
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface GigabitEthernet0/0
SHARK(config-if)#
```

☐ Top

SHARK

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/0/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 172.18.110.194

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/1
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#no shutdown
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface GigabitEthernet0/0
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/0
SHARK(config-if)#
```

☐ Top

SHARK

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/0/1

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 172.18.110.201

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
SHARK(config-if)#ip address 172.18.110.201 255.255.255.252
SHARK(config-if)#no shutdown
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface GigabitEthernet0/0
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/0
SHARK(config-if)#
SHARK(config-if)#exit
SHARK(config)#interface Serial0/0/1
SHARK(config-if)#
```

☐ Top

TUBIRON

TIBURON

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/1/0

Serial0/1/1

GigabitEthernet0/0

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00D0 BC84 3401

IP Configuration

IPv4 Address 172.18.110.129

Subnet Mask 255.255.255.192

Tx Ring Limit 10

Equivalent IOS Commands

```
TIBURON(config)#interface Serial0/1/1
TIBURON(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#ip address 172.18.110.198 255.255.255.252
TIBURON(config-if)#no shutdown
TIBURON(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

TIBURON(config-if)#exit
TIBURON(config)#interface GigabitEthernet0/0
TIBURON(config-if)#
```

☐ Top

TIBURON

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/1/0

Serial0/1/1

Serial0/1/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 172.18.110.202

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
TIBURON (config-if)#no shutdown
TIBURON (config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up
TIBURON (config-if)#exit
TIBURON (config)#interface GigabitEthernet0/0
TIBURON (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
TIBURON (config-if)#exit
TIBURON (config)#interface Serial0/1/0
TIBURON (config-if)#
```

☐ Top

TIBURON

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/1/0

Serial0/1/1

Serial0/1/1

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 172.18.110.198

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
TIBURON (config-if)#exit
TIBURON (config)#interface GigabitEthernet0/0
TIBURON (config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
TIBURON (config-if)#exit
TIBURON (config)#interface Serial0/1/0
TIBURON (config-if)#
TIBURON (config-if)#exit
TIBURON (config)#interface Serial0/1/1
TIBURON (config-if)#
```

☐ Top

PC1

PC1

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name: PC1

Interfaces: FastEthernet0

Gateway/DNS IPv4

☐ DHCP

☒ Static

Default Gateway: 172.18.110.1

DNS Server:

Gateway/DNS IPv6

☐ Automatic

☒ Static

Default Gateway:

DNS Server:

☐ Top

PC1

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status: ☐ 100 Mbps ☐ 10 Mbps ☒ On

Bandwidth: ☐ Auto

Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address: 0002.4AEA.1AA2

IP Configuration

☐ DHCP

☒ Static

IPv4 Address: 172.18.110.30

Subnet Mask: 255.255.255.224

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address: /

Link Local Address: FE80::202:4AFF:FEEA:1AA2

☐ Top

PC2

PC2

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name: PC2

Interfaces: FastEthernet0

Gateway/DNS IPv4

☐ DHCP

☒ Static

Default Gateway: 172.18.110.33

DNS Server:

Gateway/DNS IPv6

☐ Automatic

☒ Static

Default Gateway:

DNS Server:

☐ Top

PC2

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status: ☒ On

Bandwidth: ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex: ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address: 00E0.8FD8.CCD1

IP Configuration

☐ DHCP

☒ Static

IPv4 Address: 172.18.110.62

Subnet Mask: 255.255.255.224

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:8FFF:FED8:CCD1

☐ Top

PC3

PC3

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC3

Interfaces FastEthernet0

Gateway/DNS IPv4

☐ DHCP

☒ Static

Default Gateway 172.18.110.129

DNS Server

Gateway/DNS IPv6

☐ Automatic

☒ Static

Default Gateway

DNS Server

☐ Top

PC3

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0004.9A85.0832

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.18.110.190

Subnet Mask 255.255.255.192

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address FE80::204:9AFF:FE85:832

☐ Top

PC4

The screenshot shows the 'PC4' configuration window with the 'Config' tab selected. The left sidebar has a tree view with 'GLOBAL' expanded, showing 'Settings', 'Algorithm Settings', and 'INTERFACE'. Under 'INTERFACE', 'FastEthernet0' is selected. The main area is titled 'Global Settings' and contains the following fields:

- Display Name: PC4
- Interfaces: FastEthernet0 (dropdown)
- Gateway/DNS IPv4:
 - ☐ DHCP
 - ☒ Static
 - Default Gateway: 172.18.110.65
 - DNS Server: (empty)
- Gateway/DNS IPv6:
 - ☐ Automatic
 - ☒ Static
 - Default Gateway: (empty)
 - DNS Server: (empty)

At the bottom left, there is a 'Top' button.

The screenshot shows the 'PC4' configuration window with the 'Config' tab selected. The left sidebar has a tree view with 'INTERFACE' expanded, showing 'FastEthernet0' and 'Bluetooth'. 'FastEthernet0' is selected. The main area is titled 'FastEthernet0' and contains the following settings:

- Port Status: ☒ On
- Bandwidth: ☒ 100 Mbps, ☐ 10 Mbps, ☒ Auto
- Duplex: ☐ Half Duplex, ☒ Full Duplex, ☒ Auto
- MAC Address: 0000.0CAA.16C8
- IP Configuration:
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 172.18.110.126
 - Subnet Mask: 255.255.255.192
- IPv6 Configuration:
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: (empty)
 - Link Local Address: FE80::200:CFF:FEAA:16C8

At the bottom left, there is a 'Top' button.

Task 2 – Routing Table

1. Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.

- a. CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

- b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

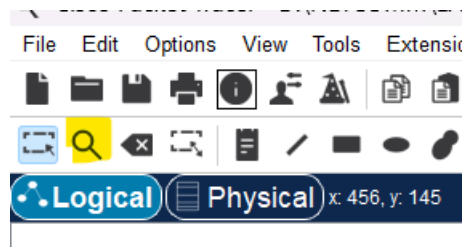


Figure B

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

JERUNG

A. CLI

```

JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

```

B. PT tool

Routing Table for JERUNG

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0

SHARK**A. CLI**

```

SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

```

B. PT tool

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0

TUBIRON

A. CLI

```
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0
```

B. PT tool

Routing Table for TIBURON					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.196/30	Serial0/1/1	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0	
C	172.18.110.200/30	Serial0/1/0	---	0/0	

2. Try to ping PC2 from PC1, paste the results here.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time=1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- Try to ping PC4 from PC1, paste the results here.

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Explain the reason(s) behind the results.
PC2 has successfully ping from PC1 because they are in same subnet and router. PC4 cannot ping from PC1 because they are in different subnet and router. Different router cannot communicate with each other.
- What needs to be done to ensure all PCs can ping each other successfully?
 - Ensure that all PCs have correct IP addresses and subnet masks configured
 - Ensure that each PC has the correct default gateway configured

Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON					Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	L	172.18.110.161/32	GigabitEthernet0/0	---	0/0

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- a. What can you say about the addresses used in the ‘network’ instructions in Figure E?
 The ‘network’ instruction is used to make RIP to include all interfaces within the subnet. So that devices in different subnets and routers can communicate with each other
3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```

TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200

```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for JERUNG					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0	R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0	C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0	L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0	L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0	C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0	L	172.18.110.202/32	Serial0/1/0	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1					

- a. What are the changes seen in TIBURON?

There is a network with type R(rip) shown in the routing table

- b. What are the Networks with type R in TIBURON and JERUNG?

TIBURON

- 172.18.110.0/27
- 172.18.110.32/27
- 172.18.110.192/30

JERUNG

- 172.18.110.128/26
- 172.18.110.200/30

- c. Ping PC3 from PC1. Was it successful?

```
C:\>ping 172.18.110.191

Pinging 172.18.110.191 with 32 bytes of data:

Reply from 172.18.110.198: bytes=32 time=1ms TTL=254
Reply from 172.18.110.198: bytes=32 time=1ms TTL=254
Reply from 172.18.110.198: bytes=32 time=1ms TTL=254
Reply from 172.18.110.198: bytes=32 time=1ms TTL=254

Ping statistics for 172.18.110.191:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

It was a success to ping PC3 from PC1.

- d. Ping PC4 from PC1. Was it successful?

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC1 failed to ping PC4.

- e. Explain the reasons for your answer.

PC1 successfully pings PC3 because both are in the subnet and are directly connected to JERUNG. JERUNG and TUBIRON have RIP connection, so that the PCs in the subnets connected to JERUNG and TUBIRON can successfully

communicate with each other because the routing information is exchanged and propagated.

PC1 failed to ping PC4 because SHARK does not have a RIP connection with JERUNG. This is the reason why PCs in the subnets connected to SHARK cannot communicate with the PCs in the subnets connected to JERUNG. SHARK router don't have configuration of RIP.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.64
SHARK(config-router)#network 172.18.110.196
SHARK(config-router)#network 172.18.110.200
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

- g. Open router SHARK's routing table and paste here.

Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	Success
2	Success
3	Success

PC1

```

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=17ms TTL=126

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 5ms

C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=14ms TTL=126
Reply from 172.18.110.30: bytes=32 time=17ms TTL=126
Reply from 172.18.110.30: bytes=32 time=32ms TTL=126
Reply from 172.18.110.30: bytes=32 time=7ms TTL=126

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 32ms, Average = 17ms

```

PC2

```

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=14ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 4ms

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=15ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=13ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 15ms, Average = 7ms

```

PC3

```

C:\>ping 172.18.110.191

Pinging 172.18.110.191 with 32 bytes of data:

Reply from 172.18.110.202: bytes=32 time=1ms TTL=254
Reply from 172.18.110.202: bytes=32 time=1ms TTL=254
Reply from 172.18.110.202: bytes=32 time=1ms TTL=254
Reply from 172.18.110.202: bytes=32 time=1ms TTL=254

Ping statistics for 172.18.110.191:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 172.18.110.191

Pinging 172.18.110.191 with 32 bytes of data:

Reply from 172.18.110.202: bytes=32 time=15ms TTL=254
Reply from 172.18.110.202: bytes=32 time=1ms TTL=254
Reply from 172.18.110.202: bytes=32 time=1ms TTL=254
Reply from 172.18.110.202: bytes=32 time=1ms TTL=254

Ping statistics for 172.18.110.191:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 15ms, Average = 4ms

```

Task 3 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: **192.168.1.0/24**
 - b. As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.254
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?

PC1

```

C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

PC4

```
C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 172.18.110.65: Destination host unreachable.
Reply from 172.18.110.65: Destination host unreachable.
Request timed out.
Reply from 172.18.110.65: Destination host unreachable.

Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

PC1 and PC4 cannot ping PC3

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for TIBURON					Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1	R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1	R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1	C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1	L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1	C	172.18.110.192/30	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0	L	172.18.110.194/32	Serial0/0/0	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0	R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/1/0	---	0/0	R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
L	172.18.110.202/32	Serial0/1/0	---	0/0	C	172.18.110.200/30	Serial0/0/1	---	0/0
C	192.168.1.0/24	GigabitEthernet0/0	---	0/0	L	172.18.110.201/32	Serial0/0/1	---	0/0
L	192.168.1.1/32	GigabitEthernet0/0	---	0/0					

- e. Referring to the routing table, explain your findings.
 The routing table of SHARK does not contain the 192.168.1.0/24 subnet. The routing table of TUBIRON contains the 192.168.1.0/24. This is the reason why PC1 and PC4 cannot ping PC3.
- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?
 Ensure all routes have the correct network in RIP to propagate 192.168.1.0/24. Update the RIP in TUBIRON and propagate the new route for other subnets.
- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```

TIBURON#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#|

```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	Success
2	Success
4	Success

PC1

```

C:\>ping 172.18.110.30

Pinging 172.18.110.30 with 32 bytes of data:

Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126
Reply from 172.18.110.30: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

```

PC2

```

C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Reply from 172.18.110.62: bytes=32 time=19ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126
Reply from 172.18.110.62: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 19ms, Average = 5ms

```

PC4

```
C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126
Reply from 172.18.110.126: bytes=32 time=25ms TTL=126
Reply from 172.18.110.126: bytes=32 time=1ms TTL=126

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 25ms, Average = 7ms
```

REFLECTION

What have you learned in this task?

This lab helped me understand subnetting and routing protocols. I learned about the configuration of IP addresses, subnet masks, and gateways to establish connectivity between devices. The importance of verifying routing configurations and effectively propagating routes was emphasized when troubleshooting connectivity issues. Using tools such as CLI and Packet Tracer enhanced my hands-on knowledge by allowing me to view and tune network topologies. Overall, this lab was an improvement in my network design and troubleshooting skills.